

Scatter-All

Scatter-All is a powerful Unity add-on designed to quickly distribute objects in the scene with configurable parameters.

Ideal for designers working on detailed environments, Scatter-All makes the asset placement process simple, fast, and customizable.

Table of Contents

1. [Introduction](#)
2. [Requirements](#)
3. [Installation](#)
4. [Main Features](#)
5. [Usage](#)
6. [Notes and Credits](#)

1. Introduction

Scatter-All is a versatile tool that allows you to:

- Precisely distribute 3D assets (e.g., rocks, trees, decorative objects) over other meshes.
- Configure parameters such as random rotation, scale, and distance between objects.
- Optimize the time needed for setting up complex scenes.

2. Requirements

- Unity 6 or higher (Custom modifications for older versions are possible upon request).
- Compatible with HDRP, URP, and Built-in pipelines.
- No dependencies on additional plugins.

3. Installation

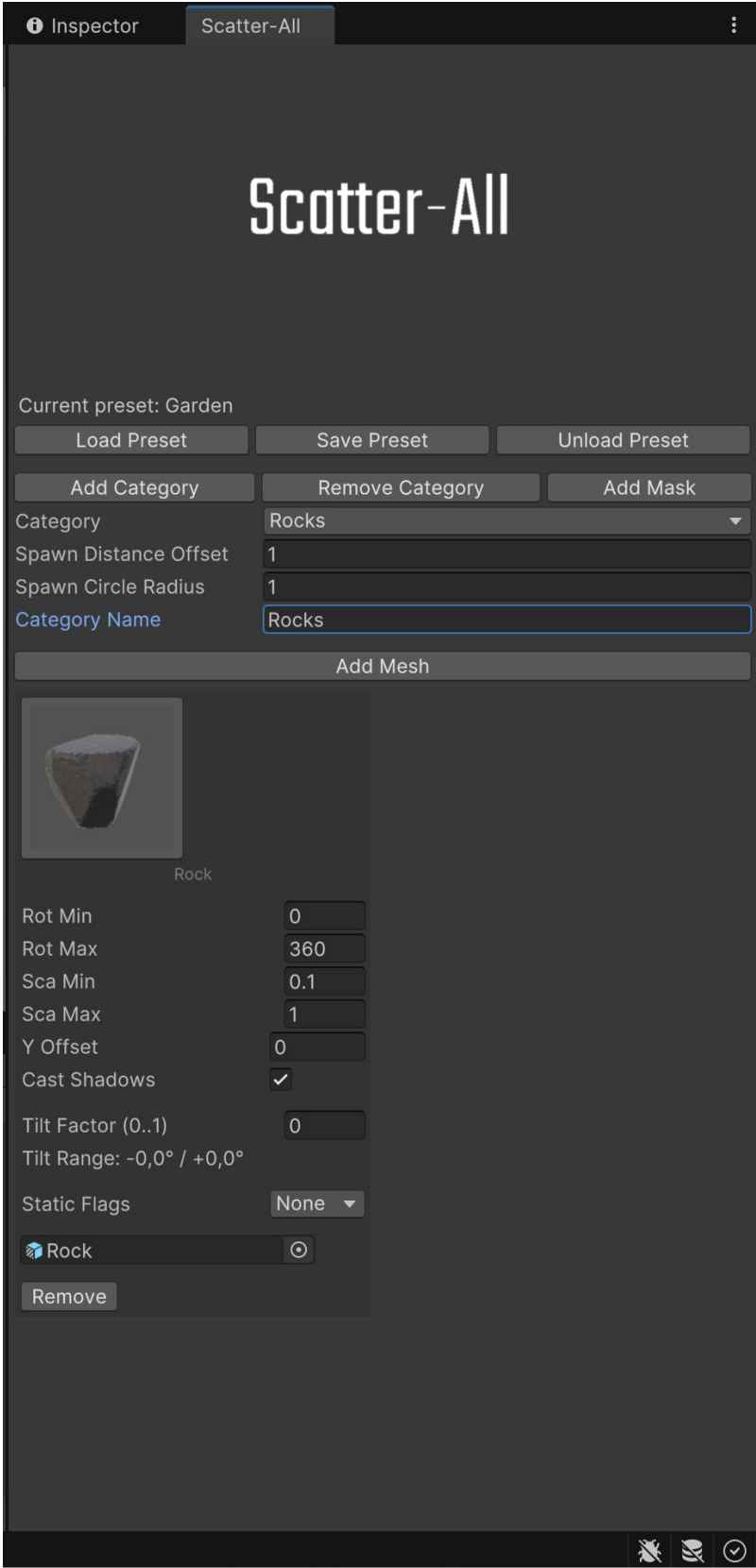
- Download the `Scatter-All.unitypackage` file.
- Import the package into Unity with a simple double-click on the package file.

Demo Scene Shaders:

- The shaders in the demo scene are set up for **URP** by default.
- If you want to try the demo scene using **Unity Standard** or **HDRP**, you need to import the respective `.package` file located in `SA_Demo/Shaders_Presets`.

This ensures the demo scene works correctly across different render pipelines.

4. Main Features



Load, Save, and Unload Preset: Enables the creation and switching of presets for better zone control.

- The **Load** button allows you to quickly load a preset. For example, you can have one preset for the garden of Home_A and another with different vegetation and objects for Home_B. With Load, you can switch between them quickly for level design or small adjustments.
- The **Save** button allows you to save your preset to any folder in your project.
- The **Unload** button removes (but does not delete) the current preset. This is particularly useful when transferring the Scatter-All folder to another project with a simple drag-and-drop. The preset might remain loaded, and opening the window could inadvertently modify an unnecessary preset, risking corruption. Unload is primarily used during development for testing across different pipelines, but it is also beneficial for end users.

Add Category, Remove Category: Create specific object categories for maximum control. You can create categories for Grass, Rocks, Trees, etc., and each category will save its “spawn per mesh” settings.

Add Mask: Excludes meshes spawned by Scatter-All. By default, the mask has four vertices, and all meshes within its boundaries will be disabled. Holding **CTRL** displays a point on the edges of the mask; clicking on the desired point adds another vertex.

- **Tip:** Always move masks using the Move Tool and by dragging the green plane (Y plane, not the arrow). You can also move mask vertices and scale them, though it's not recommended.

Spawn Distance Offset: Allows you to set the distance between meshes during spawning. Useful when spawning by holding the left mouse button.

Spawn Circle Radius: Adjust the brush resolution for optimizing the deletion of spawned meshes.

Add Mesh: Adds meshes to the category specified during **Add Category**.

- **Tip:** Always use a prefab for better control. Spawned meshes will be added automatically to the hierarchy as children of a GameObject named after their category. For instance, meshes added to the "Grass" category will appear as children of a GameObject named "Grass" in the hierarchy.

RotMin and RotMax: Allows you to set rotation values for meshes during spawning. Great for random placement. Keeping it at 0–360 is often the best option. For specific meshes, manual adjustments can be made later.

ScaMin and ScaMax: Sets the scale range for spawned meshes, adding variety to their size.

Y-Axis Control: Specifies the mesh position on the Y-axis before it's used.

Tilt Factor: Adds a slight tilt to mesh rotations, ranging from (-0.1° to 0.1°) up to (-45° to 45°). Perfect for adding variety to trees but usable on any mesh.

Cast Shadows: Prevent certain meshes from casting shadows, which can significantly improve game performance.

Static Flags: Manage your Static Flags directly from Scatter-All, streamlining your workflow.

Remove: Deletes meshes from the selected category.

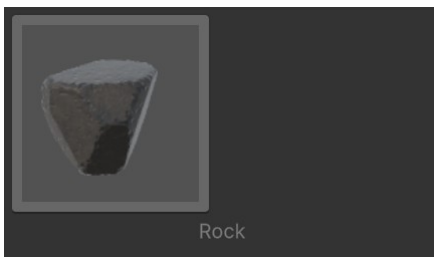
- **Note:** Already spawned meshes in the scene will remain in place. It is recommended to manually delete them from the hierarchy as well.

5. Usage

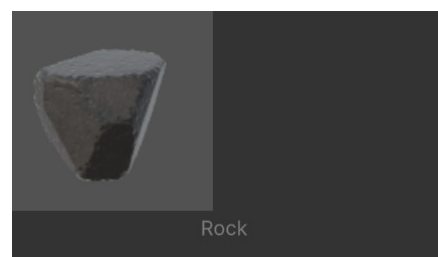
Spawn Mode

- **Activating Spawn Mode:**

Click on the preview of the mesh with the **left mouse button**. The borders of the preview will disappear, indicating that you are now in **Spawn Mode**.



Unselected Mesh



Selected Mesh

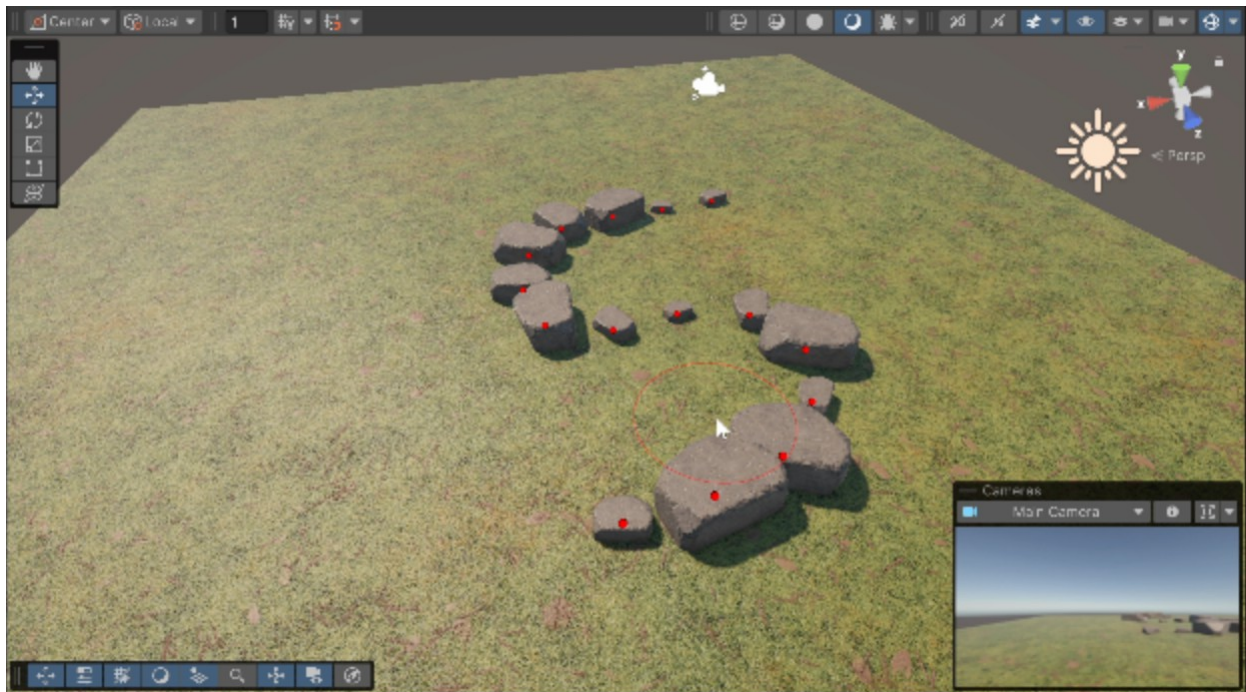
- **Spawning Meshes:**

- **Single Placement:** Click on a specific point on a target mesh to spawn an instance of the selected object.
- **Continuous Placement:** Hold down the left mouse button to spawn multiple objects continuously. The system will use the **Spawn Distance** parameter to determine the spacing between the spawned meshes.

- **Delete Mode:**

Hold **CTRL** to enter **Delete Mode**. In this mode:

- The pivot of the spawned meshes will be replaced by a red sphere, signaling that you are in Delete Mode.
- The size of the **Spawn Circle** determines the area of deletion.
- You can delete objects with a single click or by holding the left mouse button for continuous deletion.



Mask Usage

- **Adding a Mask:**

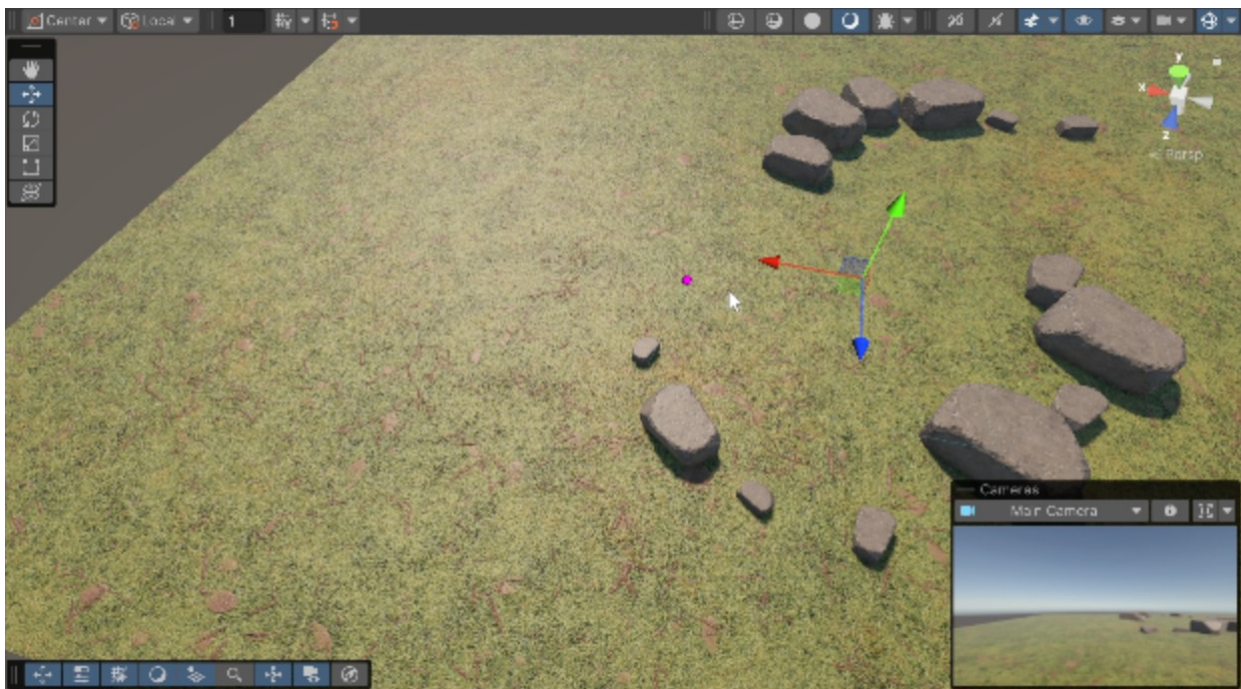
Click on the **Add Mask** button. A new mask will appear in the hierarchy, as child of ScatterAll_Masks.

- **Manipulating the Mask:**

1. Select the mask in the hierarchy.
2. Move the mask using its pivot to reposition it.
3. Adjust the vertices to expand or shrink the mask.

- **Edit Mode:**

Hold **CTRL** with the mask selected to enter **Edit Mode**. In Edit Mode, you can add new vertices to the mask for finer control. Simply click on the desired edge point to create a new vertex.



- **Effect of Masks:**

All spawned meshes within the mask area will be disabled.

1. If the mask is **deleted**, the disabled meshes will automatically re-enable themselves.

This system ensures precise control over mesh spawning and deletion while offering flexible masking capabilities for advanced customization.

6. Notes and Credits

Scatter-All is developed by Francesco Albamonte and is currently still in development. This is why the add-on is offered at a very affordable price. I already have some ideas for expanding its features, but these will take time to implement.

In the meantime, I'm available for any optimizations or updates needed to support future versions of Unity.

If you enjoy using Scatter-All and would like to support its development, please consider buying me a coffee:

ko-fi.com/bukyja

Follow Me on Social Media

Stay updated with my work and future projects by following me on social media:

- **ArtStation:** <https://bukyja.artstation.com/>
- **Instagram:** https://www.instagram.com/albamonte_francesco/
- **YouTube:** <https://www.youtube.com/@BukyjaGameDev3D>
- **Twitch:** <https://www.twitch.tv/FrankyAlba>

Please note that my content is primarily in Italian.

Thank you for your support, and see you soon!